

Optimal Stopping in Romantic Ambiguity: A Bayesian Game-Theoretic Analysis with Monte Carlo Simulation of Situationship Dynamics

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Abstract

This paper models the situationship as a coordination problem under incomplete information. In the symmetric environment, two agents may rationally coordinate on mutual vagueness even when mutual commitment is Pareto superior, because unilateral disclosure carries a private vulnerability cost that makes the vague equilibrium risk-dominant. The asymmetric extension introduces a commitment-seeking agent who updates beliefs via Bayes' rule over a partner's hidden type and initiates clarification whenever the posterior probability of reciprocal interest crosses an endogenously derived threshold. A Monte Carlo simulation over $N = 10,000$ synthetic trajectories shows that the expected stopping time under baseline calibration clusters near the Ghosting Event Horizon identified by Nahadi (2026a), implying that the moment at which commitment becomes strategically rational often arrives shortly before the arrangement loses structural integrity. The policy implication is unambiguous: one should reduce the private cost of vulnerability, not the evidentiary standard for action.

1 Introduction

The situationship occupies a recognizable place in contemporary interpersonal markets. Two agents exchange attention, time, and occasional weekend logistics while leaving the contractual status of their arrangement strategically underdefined. From the standpoint of ordinary conversation this is confusing. From the standpoint of economics it is a coordination problem with unusually bad data.

Recent formal treatments have begun to impose structure on the phenomenon. Nahadi (2026a) models the situationship as an unstable isotope whose structural integrity decays exponentially in the absence of the activation energy supplied by a commitment conversation. The contribution of that paper is temporal: it identifies the rate of deterioration, derives a half-life of roughly 4.62 weeks under the baseline calibration, and locates a Ghosting Event Horizon at 20 percent structural integrity below which the arrangement undergoes complete failure. The model has no strategic layer. Agents do not choose actions; they simply decay. Nahadi (2026b) shifts attention from structure to belief. The central contribution of that paper is a demonstration that agents who ignore clear evidence of disinterest are not failing at Bayesian inference; they are performing it correctly from a corrupted initial condition called the Delusion Prior. Given a sufficiently extreme prior,

the posterior probability of reciprocal interest can remain embarrassingly high through several instances of blatant non-responsiveness. The model is also without a strategic layer, but in a different direction: the target of inference is passive. One player updates; the other merely emits evidence.

Both papers stop one layer too early. The decay model raises the question of why agents fail to intervene before deterioration becomes terminal. The Bayesian model raises the question of what the stable outcome of bilateral inference under strategic uncertainty looks like. Neither paper asks whether the situationship itself can be a rational equilibrium, nor when a player who has already developed a preference for commitment should optimally stop waiting and initiate costly clarification.

This paper addresses that gap by layering three tools from economics and decision theory onto one modestly humiliating subject. The first layer uses a two-player coordination game to show that mutual vagueness can persist as a risk-dominant equilibrium even when mutual commitment produces strictly higher payoffs for both agents. The second layer introduces incomplete information in the sense of Harsanyi (1967): partner type is hidden, weekly signals are noisy, and both agents are choosing actions rather than simply emitting observables. This is structurally different from the single-agent updating of Nahadi (2026b). The third layer embeds the resulting belief process in an optimal stopping problem and uses Monte Carlo simulation to estimate the expected timing of the commitment conversation under a range of calibrations.

The central claim is straightforward. A situationship is not necessarily an emotional accident. It can be a stable strategic outcome produced by rational agents facing costly unilateral disclosure and poor information about the other player’s willingness to reciprocate. In that sense, its defining pathology is not irrationality but equilibrium selection. The tragedy is that the equilibrium most likely to survive under immediate play is frequently the one least capable of surviving time.

2 Related Literature

Nahadi (2026a) proposes the first formal decay law for romantic ambiguity. Structural integrity $S(t)$ evolves according to

$$\frac{dS}{dt} = -\lambda S(t), \quad S(t) = S_0 e^{-\lambda t}, \quad (1)$$

where $S_0 = 1$ denotes the initial honeymoon-phase integrity and $\lambda > 0$ captures ambient anxiety, mixed-signal frequency, and the social pressure exerted by associates who repeat the phrase “so are you guys actually dating.” Under the baseline calibration $\lambda = 0.15$, the implied half-life is

$$t_{1/2} = \frac{\ln 2}{\lambda} \approx 4.62 \text{ weeks}, \quad (2)$$

and the time at which structural integrity falls to the critical threshold $S_{\text{crit}} = 0.20$ — the Ghosting Event Horizon — is

$$t_d = \frac{\ln(S_0/S_{\text{crit}})}{\lambda} = \frac{\ln 5}{0.15} \approx 10.73 \text{ weeks}. \quad (3)$$

This is a strong reduced-form description of deterioration. It remains silent, however, on why agents fail to intervene before deterioration becomes terminal, and it contains no decision layer.

Nahadi (2026b) shifts attention to belief dynamics. Using the odds form of Bayes’ theorem,

$$O(H \mid E^{(n)}) = O(H) \prod_{i=1}^n \Lambda_i, \quad (4)$$

that paper shows that a sufficiently extreme prior — the Delusion Prior, characterized by $P(H) \rightarrow 1$ — can sustain a posterior probability of reciprocal interest above 90 percent through four instances of blatant non-response. The insight is epistemic: some agents are not bad at updating; they are bad at beginning. The model is nonetheless expressly unilateral. One player infers while the other emits evidence; strategic interaction does not enter.

The present paper extends both contributions. It formalizes mutual vagueness as a coordination equilibrium, drawing on the treatment of strategic uncertainty in Camerer (2003). It recasts the belief-formation problem as bilateral incomplete information in the sense of Harsanyi (1967, 1968a,b), where hidden types shape expected payoffs and both players choose actions simultaneously. Finally, it embeds the resulting belief process in an optimal stopping framework surveyed by Ferguson (1989), producing an estimable timing rule for the commitment conversation under noisy signals.

3 Layer I: Nash Equilibrium and the Persistence of Vagueness

3.1 Setup

Consider two players, A and B , who simultaneously choose between commitment (C) and vagueness (V). Let u denote the utility from mutual commitment and v the utility from mutual vagueness, with $0 < v < u$. Let $c > 0$ denote the vulnerability cost paid by a player who commits unilaterally and is not met with reciprocation, and let $\delta \geq 0$ denote the optionality value enjoyed by the player who remains vague while the other discloses. The stage game is as follows.

	$B : C$	$B : V$
$A : C$	(u, u)	$(-c, v + \delta)$
$A : V$	$(v + \delta, -c)$	(v, v)

Table 1: Stage game payoffs.

The restrictions $u > v + \delta$ and $v > -c$ are imposed to preserve the dual-equilibrium structure. The first ensures that mutual commitment dominates the consolation prize of keeping one’s composure after someone else has already become sincere. The second ensures that mutual vagueness is still preferable to unilateral vulnerability.

3.2 Existence of Two Equilibria

Best-response analysis is immediate. Given $B = C$, player A compares u with $v + \delta$ and chooses C because $u > v + \delta$ by assumption. Given $B = V$, player A compares v with $-c$ and chooses V because $v > -c$ by assumption. By symmetry, the same applies to player B . Therefore (C, C) and (V, V) are both Nash equilibria.

The first equilibrium is payoff-dominant: both agents obtain $u > v$. The second is defensively stable: both obtain only v , but neither pays the vulnerability cost c .

3.3 Risk Dominance

Let x denote the probability with which player B is expected to commit. Player A is indifferent between C and V when

$$xu + (1 - x)(-c) = x(v + \delta) + (1 - x)v, \quad (5)$$

which yields the critical belief

$$x^* = \frac{v + c}{u + c - \delta}. \quad (6)$$

Commitment is optimal for $x > x^*$ and vagueness is optimal for $x < x^*$. The equilibrium (V, V) is risk-dominant — in the sense that its basin of attraction is larger than one-half — when $x^* > 1/2$, i.e.,

$$v + c > u - v - \delta. \quad (7)$$

In the reduced case $\delta = 0$, this simplifies to $2v + c > u$. The economic interpretation is consistent across specifications: mutual vagueness becomes the strategically safer equilibrium whenever the private downside of sincere unilateral action is large relative to the incremental surplus from formal commitment. The conversation conventionally known as “what are we” operates as a costly signal precisely because it attempts to shift play from the risk-dominant equilibrium to the payoff-dominant one.

3.4 Worked Example

Take $u = 8$, $v = 3$, $c = 6$, and $\delta = 1$. The payoff matrix is:

	$B : C$	$B : V$
$A : C$	(8, 8)	(−6, 4)
$A : V$	(4, −6)	(3, 3)

Table 2: Numerical payoffs for the worked example.

Both (C, C) and (V, V) are Nash equilibria. The indifference threshold is

$$x^* = \frac{3 + 6}{8 + 6 - 1} = \frac{9}{13} \approx 0.692. \quad (8)$$

Player A must therefore believe with probability approximately 0.7 that B will commit before commitment is itself rational. This is a demanding threshold. Under any prior that falls below 0.692, the best response is vagueness, regardless of the fact that both players would prefer (C, C) . Even in a population in which the majority of potential partners are genuinely interested, a moderately risk-averse agent will spend a substantial amount of time treating conversational holding patterns as evidence of relational progress.

The key limitation of this layer is obvious: Nash equilibrium analysis assumes complete information about the other player’s payoffs and strategic inclinations. Real situationships are rarely so transparent. The next section replaces that assumption with uncertain partner type.

4 Layer II: Bayesian Updating Under Incomplete Information

4.1 Hidden Types

Suppose player A does not observe player B 's type. Type θ_H denotes a high-interest partner who genuinely prefers the mutual-commitment equilibrium. Type θ_L denotes a low-interest partner who prefers mutual vagueness and experiences explicit definition as an avoidable administrative burden. The prior probability that B is high-interest is

$$P(\theta_H) = p, \quad P(\theta_L) = 1 - p. \quad (9)$$

Each week player A observes a signal $s_t \in \{+, -\}$. Positive signals include reciprocal initiative, reliable follow-through on plans, and messages containing grammatically complete clauses. Negative signals include delayed responses, repeated conversational deferrals, and the observable phenomenon of being consistently unavailable for everything except social media activity. Signal quality is governed by

$$P(s_t = + | \theta_H) = q_H, \quad P(s_t = + | \theta_L) = q_L, \quad (10)$$

with $q_H > q_L$. Both types can generate superficially similar observables: high-interest partners can be temporarily occupied; low-interest partners can occasionally recall that language exists.

4.2 Posterior Beliefs

After observing signal history $s^t = (s_1, \dots, s_t)$, player A updates via Bayes' rule:

$$\mu_t \equiv P(\theta_H | s^t) = \frac{P(s_t | \theta_H) \mu_{t-1}}{P(s_t | \theta_H) \mu_{t-1} + P(s_t | \theta_L)(1 - \mu_{t-1})}. \quad (11)$$

For a positive signal this yields

$$\mu_t^+ = \frac{q_H \mu_{t-1}}{q_H \mu_{t-1} + q_L(1 - \mu_{t-1})}, \quad (12)$$

and for a negative signal,

$$\mu_t^- = \frac{(1 - q_H) \mu_{t-1}}{(1 - q_H) \mu_{t-1} + (1 - q_L)(1 - \mu_{t-1})}. \quad (13)$$

This updating rule differs from Nahadi (2026b) in a structural sense. In that paper, one agent infers and the other is passive. Here, uncertainty is strategic in the Harsanyi sense: player A uses signals to infer whether player B will reward commitment or treat it as an inconvenient escalation. In a fully bilateral extension, B would simultaneously form analogous beliefs about A . The coordination failure is not merely intrapsychic delusion; it is mutual hesitation under noisy evidence from both sides of the interaction.

4.3 Decision Threshold

Suppose player A has developed a preference for commitment. Let v_A denote the per-period utility of continuing in ambiguity; for a commitment-seeking agent, $v_A < v$ and may be negative. If A initiates the conversation immediately, the expected one-shot payoff is

$$\mu_t u + (1 - \mu_t)(-c). \quad (14)$$

If A waits one more period, the immediate payoff is v_A . Ignoring continuation value, commitment is myopically optimal when

$$\mu_t u + (1 - \mu_t)(-c) \geq v_A, \quad (15)$$

which yields the posterior threshold

$$\mu_t \geq p^* \equiv \frac{v_A + c}{u + c}. \quad (16)$$

The threshold rises with vulnerability cost c and falls with the surplus from mutual commitment u . Most importantly, it falls when v_A turns negative. Once ambiguity itself becomes painful rather than merely neutral, the evidentiary standard for initiating clarification declines. When waiting starts to feel expensive, agents become willing to ask a difficult question on less decisive evidence. The social institution of the “drunk text” is, from this vantage, an endogenous response to the decline of v_A .

4.4 Worked Example

Let $u = 8$, $c = 12$, and $v_A = 1$. Then

$$p^* = \frac{1 + 12}{8 + 12} = \frac{13}{20} = 0.65. \quad (17)$$

Take prior $p = 0.30$, $q_H = 0.65$, and $q_L = 0.35$. After one positive signal,

$$\mu_1^+ = \frac{0.65 \times 0.30}{0.65 \times 0.30 + 0.35 \times 0.70} = \frac{0.195}{0.440} \approx 0.443. \quad (18)$$

After a second consecutive positive signal,

$$\mu_2^{++} = \frac{0.65 \times 0.443}{0.65 \times 0.443 + 0.35 \times 0.557} \approx \frac{0.288}{0.483} \approx 0.596. \quad (19)$$

After a third,

$$\mu_3^{+++} \approx 0.728 > p^* = 0.65. \quad (20)$$

Three consecutive positive weeks are sufficient to trigger action under this calibration. The model is capable of producing what might loosely be called courage. It merely insists on a prior sequence of events that, in live settings, may already feel unreasonably generous.

The limitation of Layer II is the mirror image of Layer I. The updating rule tells the agent what to believe at each point in time, but it does not specify when to stop once beliefs are allowed to evolve stochastically over time and the relationship itself decays in parallel. That is the problem of the simulation layer.

5 Layer III: Monte Carlo Simulation and Optimal Stopping

5.1 The Stopping Problem

Fix player A as commitment-seeking and let player B 's type remain unknown. Weekly ambiguity yields flow utility v_A , discounted at rate $\beta \in (0, 1)$. If the commitment conversation occurs at stopping time T , the payoff to A is

$$\Pi_A(T) = \sum_{t=1}^T \beta^t v_A + u \cdot \mathbf{1}(\theta_H) - c \cdot \mathbf{1}(\theta_L). \quad (21)$$

The stopping rule is posterior-based: player A initiates the conversation in the first week for which $\mu_t \geq p^*$. If this condition is never satisfied before the Ghosting Event Horizon, the interaction is classified as fade-out. Formally,

$$T^* = \inf\{t : \mu_t \geq p^*\}, \quad (22)$$

with the convention that $T^* = t_d$ if the threshold is never crossed. This is not the most general dynamic program one could write. It is, however, a tractable and faithful representation of the most common real-world strategy: “I will say something if the signs become clear enough, and if not, I will continue gathering low-resolution evidence until the whole thing dissolves.”

5.2 Calibration and Synthetic Data

All data are synthetic and generated entirely from the simulation model. The decay constant $\lambda = 0.15$ is taken directly from Nahadi (2026a). The baseline parameters are $p = 0.30$, $u = 8$, $c = 12$, $v_A = 1$, $q_H = 0.65$, $q_L = 0.35$, and $\beta = 0.92$. The calibration is deliberately behavioral rather than clinical: large c represents strong aversion to rejection, while moderate signal quality ($q_H = 0.65$, $q_L = 0.35$) ensures that signals are informative but far from decisive. For each parameter configuration, $N = 10,000$ independent trajectories are simulated. In each trajectory, B ’s type is drawn from a Bernoulli distribution with parameter p ; weekly signals are then drawn conditionally on type; posteriors are updated recursively; and the stopping rule is applied until either the conversation occurs or the process reaches $t_d \approx 10.73$ weeks.

5.3 Baseline Results

Under the baseline calibration, the posterior threshold is $p^* = 0.65$ and Table 3 reports the outcome distribution across $N = 10,000$ trajectories.

Outcome	Probability
Mutual commitment	0.207
One-sided rejection	0.075
Fade-out	0.718
Average stopping time (unconditional)	9.14 weeks
Average stopping time (talk only)	5.08 weeks

Table 3: Outcome distribution under baseline calibration ($p = 0.30$, $u = 8$, $c = 12$, $v_A = 1$, $N = 10,000$).

The dominant outcome is non-resolution. In roughly 71.8 percent of simulated paths, the posterior never crosses p^* before the arrangement enters the decay region. The unconditional average stopping time — computed by assigning fade-out trajectories a stopping value of t_d — is 9.14 weeks, which sits 1.6 weeks before structural failure. Among the minority of trajectories in which the conversation actually occurs, the average takes place at week 5.08, well before the event horizon, but this subset represents fewer than 30 percent of all paths. Rational clarification, when it arrives, arrives just early enough to be technically on time and spiritually late.

5.4 Sensitivity Analysis

Table 4 reports expected stopping times and talk probabilities across the joint support of prior belief p and vulnerability cost c . The remaining parameters are held at baseline values.

Table 4: Expected stopping week / talk probability. Rows: vulnerability cost c . Columns: prior belief p . Cell entry: expected stopping week (unconditional) / $P(\text{talk before fade-out})$.

$c \backslash p$	0.20	0.30	0.40	0.50	0.60	0.70
4	8.3/0.36	5.3/0.62	4.9/0.68	3.3/0.83	1.0/1.00	1.0/1.00
6	9.5/0.22	7.9/0.42	4.9/0.67	3.4/0.82	3.1/0.86	1.0/1.00
8	9.5/0.22	7.8/0.42	7.4/0.49	4.5/0.72	3.1/0.85	2.8/0.89
10	9.5/0.23	9.2/0.28	7.4/0.49	4.5/0.72	4.2/0.75	2.8/0.89
12	10.1/0.15	9.2/0.27	7.4/0.49	4.5/0.72	4.2/0.75	2.8/0.88
14	10.1/0.15	9.1/0.29	7.5/0.47	7.0/0.54	4.2/0.76	2.8/0.88

Several patterns are immediate. Higher prior confidence reduces the expected stopping time. Larger vulnerability cost raises it. The northwest quadrant of the table — low p , high c — produces expected stopping times of 9.5 to 10.1 weeks and talk probabilities below 0.23. This is the region in which agents wait for better information until the relationship is no longer informative enough to be worth diagnosing.

When v_A is made negative, the posterior threshold p^* falls and the agent acts sooner. In the calibration with $p = 0.45$, $c = 10$, and $v_A = -1$, the average unconditional stopping time falls to 4.79 weeks and the talk probability rises to 0.687. This acceleration does not arise because the evidence improved. It arises because the cost of continued ambiguity became larger than the cost of being wrong. Ambiguity becoming painful lowers the action threshold faster than evidence accumulation raises the posterior.

6 Results and Figures

Three substantive results emerge from the analysis.

First, the situationship is rationalizable as a coordination equilibrium. In the symmetric environment, mutual vagueness is risk-dominant when the vulnerability cost c is sufficiently large relative to the incremental surplus from commitment. The observed persistence of ambiguity therefore need not be attributed to immaturity, poor emotional literacy, or unusual weather.

Second, incomplete information amplifies delay. Once partner type is hidden and weekly signals are noisy, the threshold p^* is difficult to reach except under favorable priors. This is structurally parallel to the finding of Nahadi (2026b) but does not reduce the phenomenon to unilateral delusion. Agents remain stuck not only because they are excessively optimistic but because the available evidence is genuinely weak relative to the cost of being wrong.

Third, the optimal stopping time aligns with the decay horizon. Under baseline and high-cost parameterizations, expected action arrives around weeks 9 to 10, while Nahadi’s structural failure point lies at week 10.73. By the time accumulated evidence supports action, the structural integrity of the arrangement is nearly exhausted.

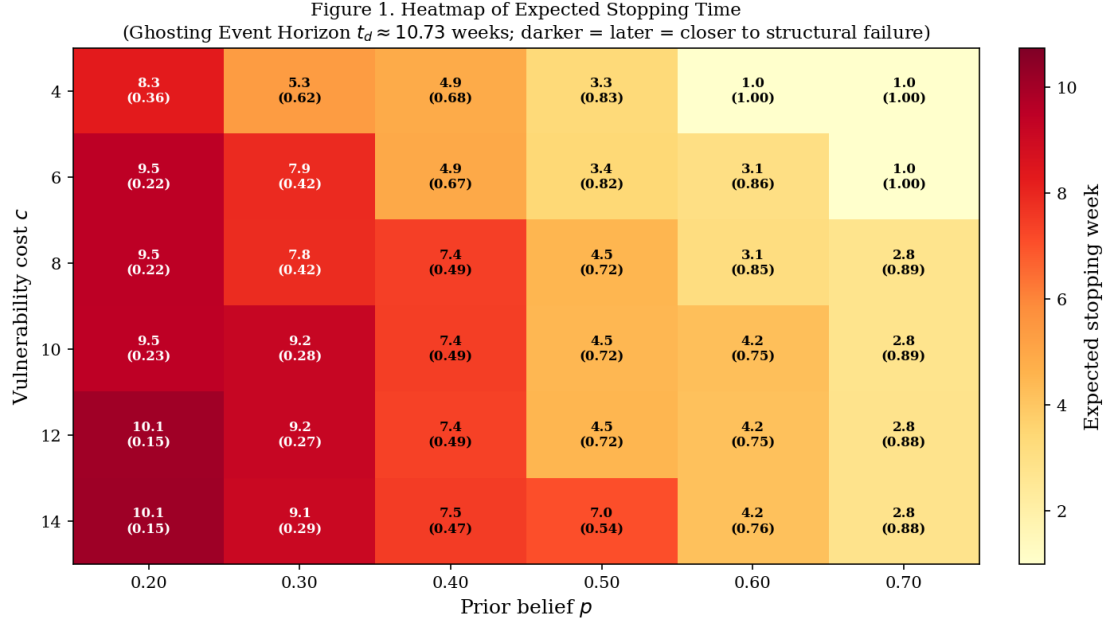


Figure 1: Heatmap of expected stopping time (unconditional) across the joint support of prior belief p (horizontal axis, 0.20–0.70) and vulnerability cost c (vertical axis, 4–14). Darker shading indicates later expected action, closer to the Ghosting Event Horizon at $t_d \approx 10.73$ weeks. Each cell reports expected stopping week / talk probability. The northwest quadrant (low prior belief, high vulnerability cost) is uniformly darkest, confirming that weak priors and large rejection costs jointly push expected action toward the boundary of structural failure.

7 Conclusion

This paper has argued that the situationship is best understood as a strategic equilibrium under incomplete information rather than as a purely emotional accident. In the symmetric game, mutual vagueness is sustained because the downside of unilateral commitment makes defensive coordination attractive. In the asymmetric extension, one player’s desire for commitment is filtered through noisy signals and a vulnerability-sensitive action threshold. The resulting stopping rule frequently delays clarification until the arrangement is already in proximity to structural breakdown.

The paper therefore provides a quantitative bridge between the two existing formal contributions. Nahadi (2026a) identifies the temporal horizon of decay. Nahadi (2026b) identifies the resilience of optimistic belief under corrupted initial conditions. The present model shows how strategic fear links the two. Under sufficiently high vulnerability cost, the optimal stopping problem converges toward non-action, and the Ghosting Event Horizon becomes the natural endpoint of equilibrium delay rather than an exogenous surprise. The decay, in other words, does not ambush the agents. It merely completes the work that fear began.

The practical implication is modest, undramatic, and will likely be received with the resistance that modest undramatic implications typically attract. One should reduce c . The relevant intervention is not to manufacture stronger priors from fragments of ambiguous textual evidence, nor to spend additional weeks collecting low-quality signals

Figure 2. Outcome Distribution
Baseline: $p = 0.30$, $u = 8$, $c = 12$, $v_A = 1$

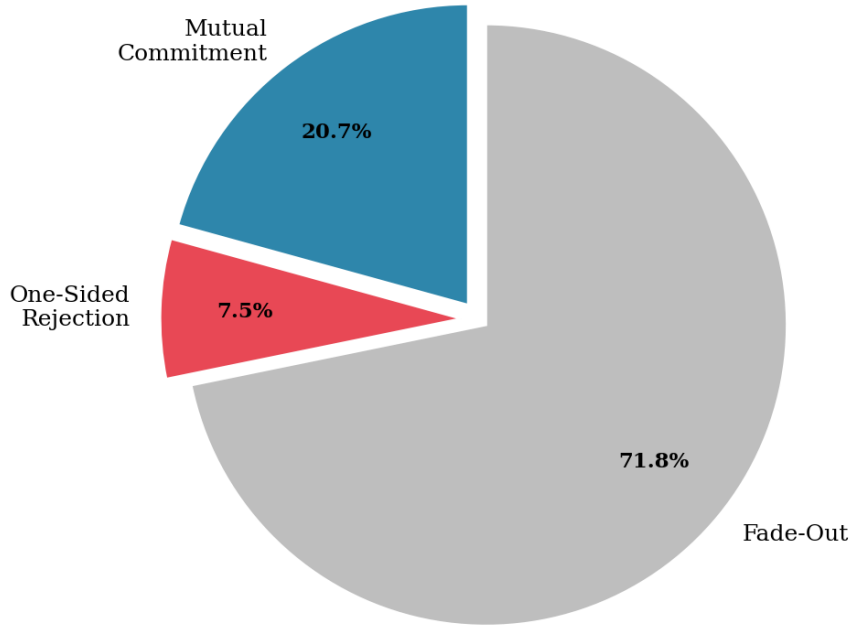


Figure 2: Outcome distribution under baseline calibration ($p = 0.30$, $u = 8$, $c = 12$, $v_A = 1$, $N = 10,000$ trajectories). Fade-out at 71.8 percent dominates the distribution. The figure is intended to look less like a love story than a hazard decomposition.

from a person whose principal observable behavior is being active elsewhere. Week 8 represents a defensible upper bound under the present calibration. Beyond that point, one is no longer patiently optimizing. One is merely allowing stochastic process notation to describe a slow administrative death.

Figure 3. Representative Posterior Trajectories
($p = 0.30, q_H = 0.65, q_L = 0.35$)

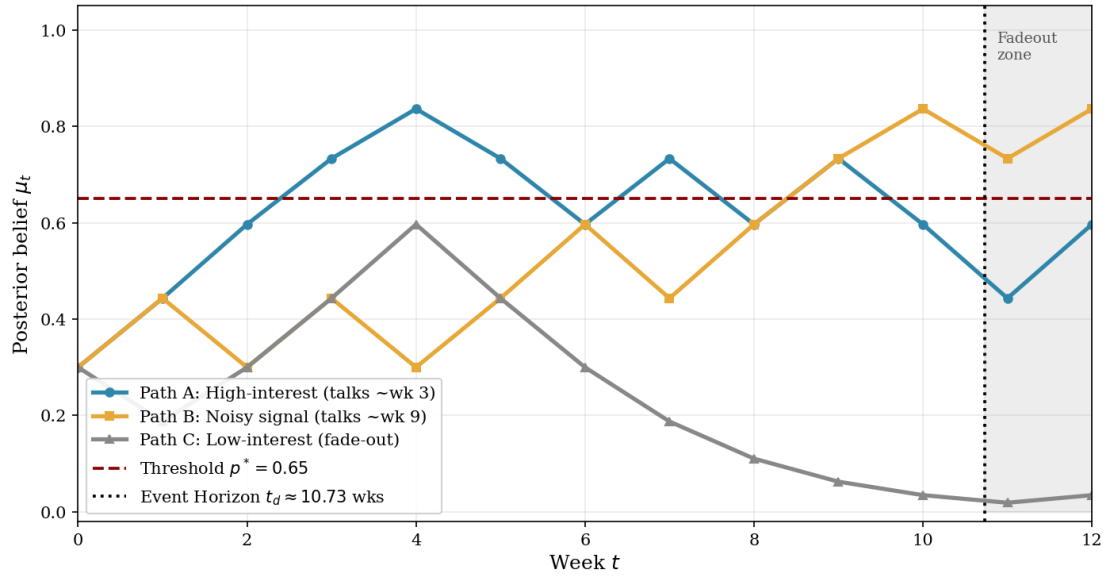


Figure 3: Three representative posterior trajectories over twelve weeks under baseline signal parameters ($q_H = 0.65, q_L = 0.35$, prior $p = 0.30$). Path A (high-interest type) crosses the threshold $p^* = 0.65$ at week 3. Path B (noisy signal realization) hovers below threshold until week 9. Path C (low-interest type) never crosses before the Ghosting Event Horizon at $t_d \approx 10.73$ weeks. The intended inference is that ambiguity persists not because the posterior fails to move, but because it moves too slowly to outrun the underlying decay process.

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